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REVIEW ARTICLE

Common causes of hair loss – clinical manifestations, trichoscopy and therapy

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Abstract

The field of hair disorders is constantly growing. The most important hair diseases are divided in non- cicatricial and cicatricial ones. Non-cicatricial alopecia are more frequent than cicatricial alopecia. The first step is to obtain a good history and physical examination. Laboratory testing is often unnecessary, while trichoscopy is fundamental for all hair diseases. Scalp biopsy is strongly suggested in cicatricial alopecia and in doubtful cases. Androgenetic alopecia, alopecia areata, telogen effluvium, trichotillomania are common causes of non- cicatricial alopecia. Frontal fibrosing alopecia, discoid lupus erythematosus, lichen planopilaris, follicullitis decalvans are some of the most common forms of cicatricial hair loss. Many treatments are available, and a prompt diagnosis is very important for the prognosis.

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Introduction

The two basic categories of alopecia are cicatricial and non-cicatricial alopecia. Non- cicatricial alopecia is more frequently seen in daily practice, while cicatricial alopecia is rare and is characterized by damaged hair follicles, skin scarring and permanent hair loss. We will deepen the most important features of these hair diseases, underlying the main differences within them (Table 1).

Non-cicatricial hair diseases

Androgenetic alopecia

Androgenetic alopecia (AGA) is the most common cause of non-cicatricial alopecia, affecting up to 50% of women and 80% of men,[1] with a frequency increasing with age after puberty. Its prevalence is higher in Caucasians than in Afro-Americans and Asians.[2,3]

AGA is characterized by a progressive hair thinning developing under the influence of a testosterone metabolite, dihydrotestosterone (DHT), against a background of genetically determined susceptibility of the hair follicles, in frontal, temporal and vertex regions.

Clinical manifestations are different in both sexes. In males AGA determines a progressive fronto-temporal recession and a vertex loss, while in women the frontal hairline is preserved and hair loss involves more or less uniformly the frontal region, posteriorly to the hairline. It's possible a female pattern in male and vice versa.

Male AGA is commonly evaluated using the Hamilton-Norwood scale that distinguishes 12 degrees of severity. In female patients the Ludwig scale, which distinguishes 3 stages, or the Sinclair 5 stages or Savin scales (6 stages) can be used.

Pull test typically shows telogen roots (roots with typically club shaped bulb), but trichoscopy is an important tool for diagnosis. The diagnostic findings are[4]: reduced hair thickness with increased number of miniaturized hair (diameter less than 0.03 mm); variation of hair diameter more than 20% in androgen-dependent regions; reduced number of hair per pilosebaceous unit (Fig. 1). In doubtful cases, a scalp biopsy can be performed. The detection of the miniaturized follicles is diagnostic, together with a decrease of terminal hairs, anagen hairs and an increase of vellus like hairs, telogen hairs and fibrous streamers. A mild to moderately dense perifollicular lymphohistiocytic inflammatory infiltrate may be seen around the infundibulum.[1]

Many topical and systemic treatments are available. Minoxidil still represents a milestone as a "hair growth stimulator", even if the precise mechanism of its action is not completely

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Table 1 Most common hair diseases

Non-cicatricial alopecia
Androgenetic alopecia
Telogen effluvium
Anagen effluvium
Alopecia areata
Trichotillomania
Loose anagen hair syndrome
Cicatricial alopecias
Lymphocyte- associated
Lichen planopilaris
Frontal fibrosing alopecia
Discoid lupus erythematosus
Central centrifugal cicatricial alopecia
Keratosis follicularis spinulosa decalvans
Neutrophil – associated
Folliculitis decalvans
Dissecting cellulitis
Mixed inflammatory infiltrate
Erosive pustular dermatitis of the scalp

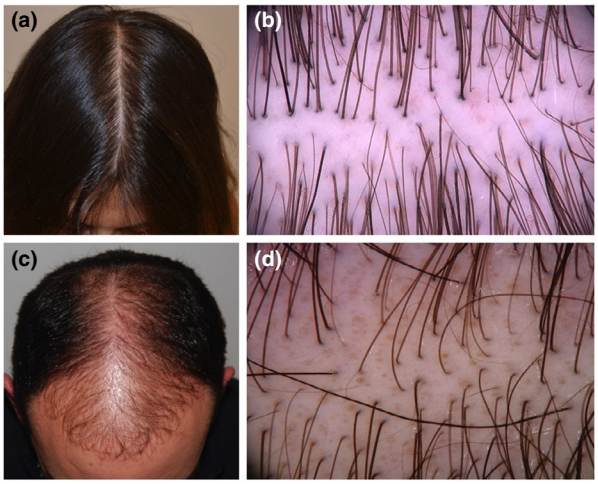


Figure 1 Clinical (a) and trichoscopic picture (b) of a case of female AGA; Clinical (c) and trichoscopic picture (d) of a case of male AGA.

understood.[5] The dosage for male AGA is 1 mL at 5% solution applied twice daily on androgen – dependent scalp areas; for female AGA 1 mL at 2% solution twice daily or 5% solution once daily. Treatment should be continuous and not suspended to maintain efficacy. Oral minoxidil represents an important field of research.[6-10] Prostaglandin analogs (PGAs), such as latanoprost, and bimatoprost can be used topically for stimulating hair growth, with good clinical response,[11] but less efficacy than minoxidil. Among systemic treatment alpha-reductase inhibitors should be mentioned.

For male AGA, the intake of finasteride 1 mg daily led to a significant increase in total hair counts, and it can be prescribed only if the beard is completely formed.[12] Administration should be protracted over time. Finasteride 1 mg and 5% topical minoxidil are the best combination therapy in male AGA and led to considerable improvement better than monotherapy.

The most common side effects are on sexual sphere: loss of libido, erectile dysfunction, and problems with ejaculation,[12-13] but they are rare and usually disappear with the withdrawal of the drug.

In female AGA, oral finasteride should only be prescribed in postmenopausal women or in women willing to use oral contraceptives, at an estimated effective dosage of 5 mg. Topical finasteride represents a very interesting field in hair research.[14-17] Dutasteride inhibits both types I and II 5-reductase and with a dosage of 0.5 mg daily has a clear evidence of its validity in male AGA, although the prescription is still off-label for AGA.[18]

There is limited data that hormones, such as oral cyproterone acetate (CPA), may be helpful in women with AGA and hyperandrogenism.[19] We suggest a strict collaboration with endocrinologist or gynaecologists to manage these therapies.

CPA acts by interfering with the binding of 5-alpha DHT to the androgen receptor and by inhibiting the secretion of FSH and LH, as the result of its progestinic action. Contemporary administration of oestrogen (ethynyl estradiol or oral contraceptives) is mandatory, to enhance the anti-androgen activity. Spironolactone can arrest hair loss progression with a favourable long-term safety profile, too.[20-22] Flutamide, a nonsteroidal selective anti-androgen, inhibits the binding of androgens to their receptors and it may be a treatment option in patients with normal androgen levels, even if a standardized dosage has not been established and it is not available for this purpose in some countries.[23,24] Other options for the treatment are: platelet-rich plasma (PRP),[25,26] low-level laser (light) therapy (LLLT)[27] and surgery.[28,29] Among these option LLLT is a promising monotherapy and an effective alternative for patients unwilling to use medical therapy or undergo surgical options.[27]

Telogen effluvium

The term telogen effluvium (TE) defines a diffuse hair loss that occurs around 3 months after a triggering event, lasting for about 6 months. In TE, hair loss is usually less than 50% of the scalp hair.[30,31] TE is more frequently in adult females and can be the consequence of 5 different situations[32]:

1 Interruption of the follicular cycle with sudden shift from the growth (anagen) phase to the rest (telogen) phase. This is the most common variety.
2 Synchronization of the follicular cycle due to elongation of anagen length.
3 Synchronization of the cycle due to shortening of the anagen phase.
4 Premature detachment of telogen hair.

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5 Increased duration of telogen phase.

TE is distinguished in acute and chronic. Acute TE results from noxious events that precipitate the entry of a large number of follicles into telogen phase. Possible causes include systemic diseases, drugs, fever, stress, weight loss, delivery, iron deficiency and inflammatory scalp disorders. Although a large number of drugs have been occasionally reported to produce TE, only for a few drugs the relation has been proven[33] and severity of hair loss depends on the drug, its dosage and patient's susceptibility.

Chronic TE is characterized by increased but mild hair shedding lasting for more than six months. This condition mostly affects middle-aged women and frequently remains unexplained. Scalp paraesthesia (trichodynia) is frequently reported.[34] The physical examination can be normal, but an accurate clinical history allows in many cases to identify the trigger event, that has to be searched around 3 months before the beginning of clinical manifestations. The most important differential diagnosis is androgenetic alopecia. Laboratory tests, as complete blood count, urine analysis, serum ferritin and T3, T4, thyroid stimulating hormone (TSH), should be performed.[35] The pull test is positive, with more than 10 telogen hair roots extracted on the same day after shampooing. There are no specific trichoscopy findings in TE. However, presence of upright regrowing hairs and predominance of hair follicle openings with only one emerging hair shaft may be indicative of telogen effluvium in absence of other causes of hair loss (Fig. 2). According to Rudnicka et al.[36] TE can be defined a diagnosis of exclusion.

In doubtful cases, a scalp biopsy can be performed: the number and density of hair follicles is usually normal, but there is an increased percentage of the hair follicles that are in the catagen or the telogen phase. If 25% of the follicles are in telogen phase, the diagnosis of telogen effluvium is confirmed. The percentage of telogen hairs should not typically be higher than 50%.[37]

The most important aspect in the management of TE is counselling the patient about the natural history and the benign course. Drugs like beta-blockers, retinoids, anticoagulants, or antithyroid need to be substitute if possible. Endocrine disorders and nutritional deficiencies should also be corrected. Assuming the lack of evidence for a specific treatment for TE, oral intake of nutritional containing iron, vitamins and aminoacids and topical application of cosmetic lotions formulated for blocking acute hair shedding, promoting hair growth,[38] including Insulin-like growth factor 1 (IGF-1), Fibroblast growth factor (FGF) and Vascular Endothelia Growth Factor (VEGF), may improve this condition. Topical or oral minoxidil has theoretical benefit in prolonging anagen phase[39-40] and can be an option. Adjuvant and newer treatments include physical therapies such as microneedling.[41]

Anagen effluvium

Anagen effluvium is a consequence of pathogenic noxae which cause an abrupt arrest of mitotic activity of matrix cells. The follicles with the highest mitotic activity undergo the strongest injury and will lead to anagen effluvium, while the follicles with less intense mitotic activity react by entering a resting phase, with fall of telogen hairs some months later. No pathogenic noxa causes a "pure" anagen effluvium, but the common causes are cytotoxic drugs, tallium, mercury and arsenic poisoning and alopecia areata.

Nowadays, anagen effluvium is often referred to chemotherapy, antimetabolites, alkylating agents, and mitotic inhibitors. Shedding usually begins abruptly, causing a diffuse and severe alopecia. Pull test shows dystrophic anagen hairs with pigmented hair shafts. Trichoscopy in the first phase reveals black dots and broken hairs, while at the end of chemotherapy besides the features described above, numerous thin hairs in regrowth, rare terminal hair, scattered black dots and circle hair are detected (Fig. 3).[42] No treatment appears to be effective preventing or stopping the hair loss. Scalp cooling, during the administration of the chemotherapy, probably decreases the local concentration and cellular uptake of drugs thanks to vasoconstriction,[43-44] with best results for alopecia induced by doxorubicin, epirubicin, and docetaxel. Scalp cooling is not recommended in haematological malignancies because of risk of scalp skin metastasis.[45] After chemotherapy, the goal is to accelerate hair regrowth: the most promising agent is minoxidil. Hair regrowth after chemotherapy, in more than 60% of patients, is often characterized by changes in hair colour, texture, shape, with darker and curler hair compared to their original form.[46]

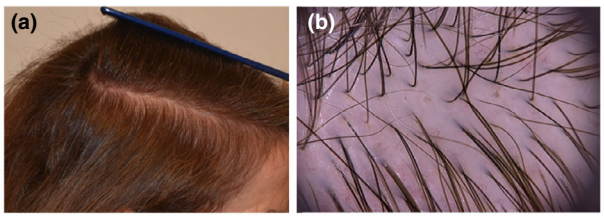


Figure 2 Clinical (a) and trichoscopic picture (b) of an acute telogen effluvium, more evident in the temporal area.

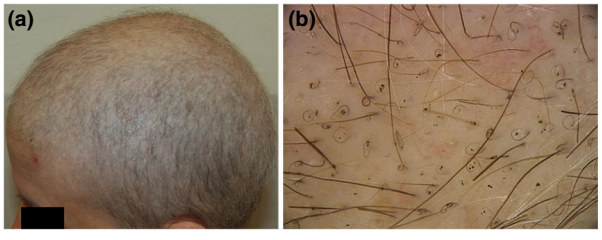


Figure 3 Clinical (a) and trichoscopic picture (b) of a chemotherapy-induced anagen effluvium.

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Alopecia areata

Alopecia areata (AA) is an autoimmune non-cicatricial alopecia. Prevalence of AA has been estimated to about 0.2% of the general population and approximately 1.7–2.1% of the population experiences an episode of aa during their lifetime.[47] The higher prevalence is between 10 and 25 years (60%) and the peak of incidence decrease with decades of life.[48]

The aetiology of AA is unknown, but is postulated that a triggering factor, still unclear, is able to induce the autoimmune process in genetically predisposed subjects. Environmental (infections, toxins), immunologic (thyroid disease, type 1 diabetes mellitus, rheumatoid arthritis, vitiligo) and genetic factors are responsible for its development.[49,50] In addition to this, it has been hypothesized that AA is caused developing antibodies that affect the structure of their own hair follicles in the anagen phase, specifically, CD4+ and CD8+ T lymphocytes infiltrate the bulb of the hair follicle.

The disease begins acutely with the appearance of one or more round patches of alopecia. Patches may involve any body area, but are more frequently localized on the scalp and the beard. On the basis of clinical extension, alopecia areata can be divided into: single or multiple patchy alopecia areata, alopecia areata of the whole scalp (alopecia totalis) and alopecia areata of the whole body hair (alopecia universalis). Rarely, AA may involve only eyelashes and eyebrows (Fig. 4). Nail can also be affected with a "geometrical" pitting.[51]

The severity of AA may be assessed by the Severity of Alopecia Tool (SALT) and graded according to the guidelines of the National Alopecia Areata Foundation.[52] Laboratory tests are always indicated to exclude autoimmune conditions. Histological examination is rarely performed, showing cell aggregates around the hair bulb, which are responsible for the inflammatory processes.[53]

AA is an anagen effluvium; with pull test and trichoscopy is possible to estimate the activity of the disease and monitor the response to treatments. On the edge of active patches there are the typical exclamation point hairs, which are pathognomonic of the disease.[48] Other typical signs are also black dots (acute phase) and yellow dots, present especially in chronic AA.

Alopecia areata may also presents with diffuse hairloss, without typical patches, in two entities: alopecia areata incognita (AAI) or diffuse alopecia areata (DAA). With trichoscopy the most frequent pattern is characterized by empty yellow dots, yellow dots containing vellus hair and small hair in regrowth. The presence of pigtail hair is very frequent and almost exclusive of AAI, while in DAA, dystrophic hair and black dots are more common, indicating the acute and deep inflammatory damage (Fig. 5). Pathology defines the diagnosis.[54]

The course of AA is unpredictable. Cases of AA which involve less than 40% of the scalp, often heal spontaneously but the course is typically relapsing and relapses could be worse than the first episode. Negative prognostic factor are childhood onset and association with atopy.

The choice of treatment, according recent guidelines,[47] depends on the extension, activity of the disease and age of the patient, and it is resumed in Table 2. It includes topical, intralesional or systemic steroids, and topical immunotherapy. Biologic drugs include anti- tumour necrosis factor (TNF)-α inhibitors, such as infliximab and etanercept, or monoclonal antibody such as efalizumab and alefacept, which until now are under investigation and seem to be less effective. A new class of molecules, the Janus kinase (JAK) inhibitors, appear to be effective, in particular tofacitinib, administered 5 to 10 mg twice daily, and oral ruxolitinib at the dose of 20 mg. Topical preparation of biologic drugs (tofacitinib 2% ointment, ruxolitinib 1% ointment) are under investigation, too. More studies are needed to

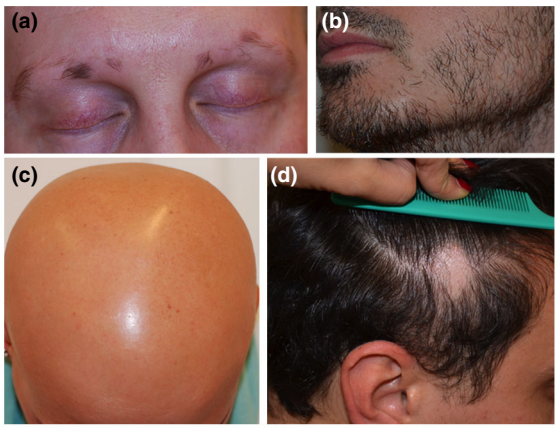


Figure 4 Different types of alopecia areata: (a) eyebrows; (b) beard; (c) totalis; (d) patchy.

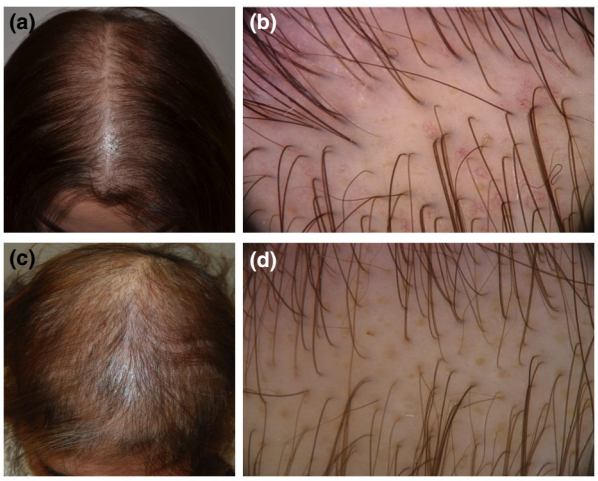


Figure 5 Clinical (a) and trichoscopic picture (b) of alopecia areata incognita; clinical (c) and trichoscopic picture (d) of diffuse alopecia areata.

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Table 2 Algorithm for the management of alopecia areata. DPCP: diphenylcyclopropenone; SADBE: squaric acid dibutyl ester

		First-line therapy		Second-line Therapy
		Acute	**Chronic**	
Children < 10 years		Potent topical corticosteroids	Topical immunotherapy (DPCP, SADBE)	
		Topical anthralin		
		Topical minoxidil	Topical anthralin	
Adults	<25% of the scalp	Intralesional corticosteroids	Topical immunotherapy (DPCP, SADBE)	
		Potent topical corticosteroids	±Topical anthralin	
			Topical minoxidil	Methotrexate Azathioprine Biologic drugs
	>25% of the scalp	Systemic glucocorticoids	Topical immunotherapy (DPCP, SADBE)	
		Potent topical corticosteroids		
			Potent topical corticosteroids	

prove effectiveness and safety of all these drugs. Adjuvant psychotherapy is useful either in children or adults affected by severe alopecia areata.

Trichotillomania

Trichotillomania is characterized by the repetitive pulling out of one's own hair leading to hair loss and functional impairment. It has a prevalence estimated of 0.5% to 2.0%.[55] It's one of the most frequent causes of transitory alopecia in children, while in adult is often present in psychiatric patients.

Trichotillomania may involve any area of the body, but more often the scalp, followed by eyebrows and pubic region. Clinically it manifests with alopecic patches of variable size and bizarre shape. Patches are not completely bald, but appear irregularly covered by truncated and broken hair.

Hair surrounding the patch are often damaged, crusts and small wounds due to scratching may be present too. In old and severe cases trichotillomania may lead to a cicatricial alopecia.

Some patients eat hair after pulling it out ("trichophagia"), which can result in gastrointestinal obstruction due to "trichobezoars", requiring surgery in extreme cases.[55]

Diagnosis of trichotillomania is often not accepted by the patients or their relatives who deny the possibility of self-induced alopecia. The pull test is usually negative.

Trichoscopy reveals abnormalities resulting from the stretching and fracture of hair shafts. Fractures may occur at different distances, resulting in irregular black dots, coiled or hook hairs, and hairs with fraying or split ends, tulip hair, flame hair, V-sign, hair powder[56-57] (Fig. 6).

The most important differential diagnosis is with alopecia areata, and, in doubtful cases, a scalp biopsy is necessary because both diseases can be associated.

Behavioural therapy can be necessary, and it used generally the habit reversal therapy (HRT).

Pharmacologic medications include a modulator of the glutamate system, the N-acetylcysteine, which has demonstrated to be useful in the control of compulsive behaviour in some patients.

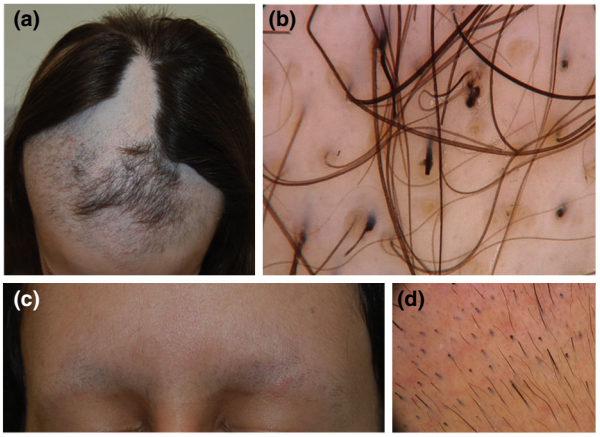


Figure 6 Clinical (a) and trichoscopic picture (b) of trichotillomania of the scalp; clinical (c) and trichoscopic picture (d) of trichotillomania of eyebrows. Both cases are children.

Clomipramine or olanzapine may also help controlling the symptoms.[58]

Loose anagen hair syndrome

Loose anagen hair syndrome (LAHS) is characterized by a defective anchorage of the hair to the follicle. It can be familial or sporadic. In some cases, the mutation of the genes that codify for adhesion's keratin (K6HF, K6IRS) specific for inner sheath[59] has been reported.

Patients, mostly female blonde children, have a typical history of not-growing or easily falling hair.

It is characterized by anagen hairs that are painlessly shed with the pull test, a ruffled proximal cuticle, absent inner root sheath, and a bent matrix.

Microscopic appearance of hair is diagnostic as roots are in anagen without sheaths, but it's positive if there is at least 10 loose anagen hair. Trichogram with the presence of 70% of loose

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anagen hair is diagnostic. LAHS appears at 2–6 years and tends to improve in adulthood, when hair becomes longer and denser. There is no current evidence for benefit from biotin supplementation,[60] but it is a good option. Topical minoxidil has shown significant clinical improvement with no adverse effects.[61]

Cicatricial hair diseases

In cicatricial alopecias permanent hair loss results from replacement of follicles by fibrosis. Hair loss may occur as a primary event due to inflammation targeted directly at the hair follicle or secondary, by exogenous factors or other scalp diseases like sarcoidosis. In this text, we focus on primary cicatricial ones.

The inflammatory infiltrate typically involves the permanent portion of the hair follicle where is present the bulge region that contains the follicular stem cell reservoir. They are classified according to a consensus-issued classification scheme based on the predominant cell type present[62]: lymphocytic, neutrophilic, or mixed. Trichoscopy can be used for an initial differential diagnosis and to select the best site for scalp biopsy, which is recommended in the majority of cases. The treatment aims at preserving the available hair and avoiding the progression.

Lymphocyte- associated

Lichen planopilaris (LPP) Lichen planopilaris (LPP) is a primary lymphocytic cicatricial alopecia defined as a follicular form of lichen planus. The aetiology is unknown, but an autoimmune pathogenesis has been postulated.

The American Hair Research Society[62] divides it into 3 clinical variants, mainly distinguished by the clinical pattern of hair loss: classical LPP, FFA, and Graham-Little-Piccardi-Lassueur syndrome.[63] Less common subtypes include FAPD (Fibrosing alopecia in pattern distribution) and CPHL (cicatricial pattern hair loss).[64]

FAPD is a progressive cicatricial alopecia histopathologically indistinguishable from LPP but limited to the area of androgen-sensitive hair follicles, showing perifollicular erythema, follicular keratosis and loss of follicular orifices in the central scalp. CPHL is a variety of lymphocytic cicatricial alopecia in a female pattern hair loss without clinically evident inflammatory signs but characterized by small pencil eraser-sized areas of focal atrichia. LPP can therefore present itself not only as a localized disease with defined patches of cicatricial alopecia but also with a diffuse involvement of the scalp (LPPDP) (Fig. 7). These diffuse types are often misdiagnosed as seborrhoeic dermatitis or androgenetic alopecia, with consequent delay in the diagnosis and a progression of the irreversible fibrosis.

Typical LPP is characterized by one or multiple areas of alopecia with irregular shape and margins, frequently localized in the central scalp with intense itching. Often yet with naked eye is possible to observe inflammatory manifestations together with skin atrophy.

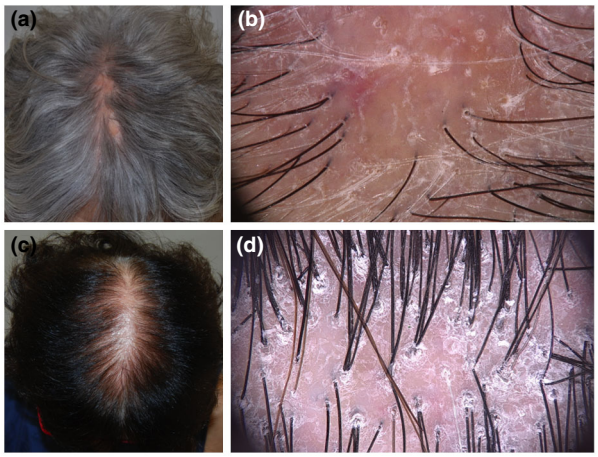


Figure 7 Clinical (a) and trichoscopic picture (b) of typical LPP; clinical (c) and trichoscopic picture (d) of diffuse LPP (LPPDP).

Physical examination of the oral mucosa, skin and nail may highlight typical manifestations in other areas. Pull test in active phase is positive with dystrophic anagen. With trichoscopy, perifollicular erythematous papules and acuminate hyperkeratotic follicular spines are seen together with absence of follicular ostia. Trichoscopic features may be similar to other cicatricial hair diseases, therefore histology is mandatory with a biopsy taken from an active area at the periphery of the patch, avoiding the scar. LPP is characterized by a lymphocyte-mediated lichenoid band–like interface dermatitis involving the follicle and interfollicular epidermis.[65]

Lichen planopilaris is very resistant to treatment and in many patients the disease progresses towards a severe cicatricial alopecia despite treatments. In mild to moderate LPP, if scalp involvement is less than 10%, intralesional steroids can be an option. Systemic corticosteroids are reserved for rapidly progressing disease and severe subjective symptoms. In patients with more than 10% scalp involvement and/or little responsiveness to topical or intralesional corticosteroids, oral hydroxychloroquine (400 mg/day) is recommended after an ophthalmologic examination. Cyclosporine has been reported to be effective in LPP unresponsive to hydroxychloroquine and corticosteroids.[66] Topical treatments with high potent corticosteroids can be used together with the other therapies. Topical minoxidil can be prescribed to increase hair diameter on the scalp, in order also to cover the affected areas.

Frontal Fibrosing Alopecia (FFA) Frontal fibrosing alopecia (FFA) affects predominantly in post/ menopausal women, even if a male presentation is also described.[67,68]

FFA is characterized by a progressive recession of frontal-temporal hairline and, in 50–75% of cases, partial or complete loss of eyebrows. The progression of the disease is generally slow and

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its response to treatment is poor. Non-inflammatory facial papules are also detected,[69] and symptoms like scalp itching and trichodynia are typical.

Pull test at the new hairline site if positive shows anagen roots with sheaths or telogen hairs. Trichoscopy is characterized by empty follicles/yellow dots, absence of follicular ostia, mild follicular hyperkeratosis, perifollicular erythema and presence of 'lonely hair' (Fig. 8).

Pathology reveals the inflammatory lymphocytic infiltrate at the level of the isthmus and infundibulum of the intermediate and terminal hair follicles, with lichenoid appearance, and perifollicular and lamellar fibrosis with loss of sebaceous glands. In our experience steroids are the treatment of choice for FFA in patients with subjective and trichoscopy signs of progression, as itching, follicular hyperkeratosis and erythema, and we suggested systemic steroids with triamcinolone acetonide for the acute phase, while a local treatment can be done with clobetasol propionate or with triamcinolone acetonide intralesional injections, followed by a maintenance therapy with 5 alpha-reductase inhibitors or hydroxychloroquine.

Discoid lupus erythematosus (DLE) Discoid lupus erythematosus (DLE) is a chronic, cicatricial, photosensitive dermatosis. DLE may occur at any age, but it most often occurs in persons aged 20–40 years.[70]

The scalp can be affected, showing usually one or few erythematous patches of pink to vivid red colour with skin atrophy, loss of follicular ostia, follicular plugging and adherent scales.

Trichoscopic features depend on the stage of the lesions: active lesions include yellow-brown dots and red dots, whereas longstanding inactive lesions show absence of follicular openings, cicatricial milky red or white patches, structureless white and brown areas, and thick arborizing vessels (Fig. 9).

The diagnosis can be confirmed by pathology and direct immunofluorescence. The histopathologic features of DLE are characterized by: an interface dermatitis with basal vacuolization, dyskeratotic or apoptotic keratinocytes, cytoid bodies, and pigment incontinence; epidermal atrophy; ortho-hyperkeratosis with areas of parakeratosis and follicular plugging; an inflammatory infiltrate composed predominantly of lymphocytes with admixed plasma cells distributed around the superficial and deeper dermal vasculature as well as adnexal structures and scattered within the interstitium.[65] Direct immunofluorescence examination if positive shows immunoglobulin G, immunoglobulin M, and C3 reactivity at the basement membrane in a continuous granular pattern.

DLE treatment of the scalp includes sunscreens, intralesional/topical/systemic steroids on the basis of the stage of diseases, and antimalarials, at dosages which are useful for cutaneous DLE.

Central centrifugal cicatricial alopecia (CCCA) CCCA is a cicatricial alopecia that primarily affects African women,[71] usually in the second or third decade of their life. Patients may complain of itching or pain.

It begins in the central area of the scalp and has a progressive and symmetric centrifuge evolution. In early stages, thinning dominates over the central scalp; in advanced stages, the affected area appears shiny and smooth with visible loss of the hair follicles.

Trichoscopy shows peripilar grey/white halo that is a specific and sensitive; honeycomb-pigmented network, pinpoint white dots (numerous white interfollicular macules irregularly distributed), hair shaft variability, white patches, perifollicular erythema, concentric white perifollicular and interfollicular scales, broken hairs, interfollicular-pigmented asterisk-like or stellate brown macules (Fig. 10).[71] Histopathological features associated with CCCA include premature desquamation of the inner root sheath (IRS), compound follicular structures with perifollicular

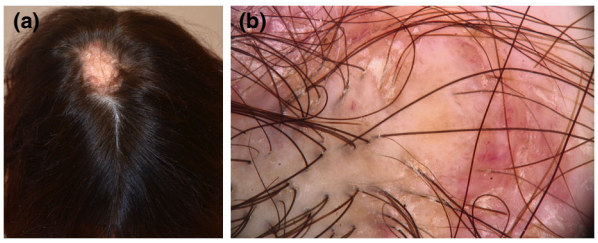


Figure 9 Clinical (a) and trichoscopic picture (b) of a patch of LED.

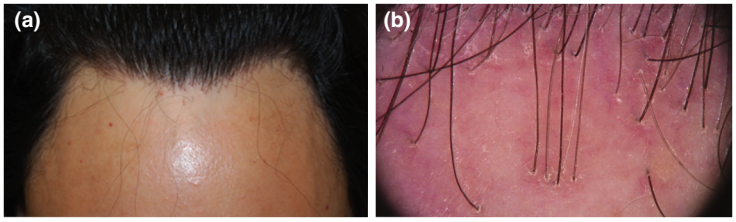


Figure 8 Clinical (a) and trichoscopic picture (b) of frontal fibrosing alopecia.

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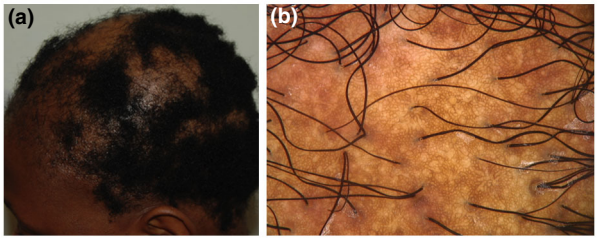


Figure 10 Clinical (a) and trichoscopic picture (b) of central centrifugal cicatricial alopecia.

fibrosis and naked hair shafts with a foreign-body inflammatory reaction in fibrous streamers.[72]

The first treatment includes suggesting to avoid hair styling procedures of these patients. Steroids, such as clobetasol propionate, are usually the first line together with topical minoxidil.

Keratosis follicularis spinulosa decalvans (KFSD) Keratosis follicularis spinulosa decalvans is a rare, X-linked, hereditary disorder of keratinization,

characterized by diffuse follicular hyperkeratosis, progressive cicatricial alopecia of scalp, eyebrows, and eyelashes and may be accompanied by photophobia, corneal and conjunctival inflammation and dystrophy, tooth abnormalities, ichthyosiform xerosis, facial erythema, atopic dermatitis, hair disorders and nail dystrophies.[73] The gene has been mapped to Xp21.2-p22.2.[74]

This condition mainly affects males with females being carriers and will have milder symptoms.

Usually KFSD begins during childhood with the appearance of follicular keratotic papules on the face, trunk and limbs. The scalp is involved later with follicular erythematous papule, nonpurulent inflammation, cicatricial hair loss, and/or atrophic depression similar to pitted scars are visible in the affected scalp areas.

After the examination of all body, scalp biopsy is mandatory, showing follicular plug with compact hyperkeratosis and hypergranulosis (initiating defect); perifollicular lymphocytic infiltrate associated with the upper follicle; concentric lamellar fibrosis, hairs haft granulomas, and scarred follicular tract remnants. Distinguishing features are peri and intrafollicular spongiosis and neutrophilic pustule formation in acute lesions.[65]

There is no specific treatment for this disorder. However, drugs such as isotretinoin and dapsone have been tried; emollients, topical corticosteroids and keratolytic agents can be also used. Genetic counselling is needed in some cases.[75]

Neutrophil – associated

Folliculitis decalvans This condition with chronic course is characterized by relapsing of follicular pustular lesions. It is also called *tufted folliculitis* since a common feature is the presence of 10-15 hairs emerging from a single follicular opening. It predominantly affects young and middle-aged male adults. The aetiology is not fully understood, even if *S. aureus* seems to play an important role.

Folliculitis decalvans (FD) involves the vertex and occipital area of the scalp, often beginning with a patch of hair loss with pustules, crusts and tufted hairs, which are pathognomonic of the disease. Bleeding, pain, burning sensation can be referred by the patient.[76]

Trichoscopic features change according to the inflammatory activity, and they are tufted hairs, perifollicular erythema, perifollicular haemorrhages, follicular pustules, yellow tubular scaling, yellow crusts (Fig. 10).[77] In most of cases, diagnosis is due with trichoscopy, but in doubtful ones, a biopsy of an active lesion can be performed (Fig. 11).

Characteristic histologic findings are: follicular plugging and intra/perifollicular neutrophilic infiltrate of the upper and middle portions of the follicle; follicular rupture with recruitment of lymphocytes, histiocytes, and plasmacells around the follicle; neutrophilic abscesses; fibrous tracts. The main distinguishing features of FD are the presence of plasmacells in advanced stages, early superficial wedge-shaped scar with loss of elastic fibre staining and the Gram stain positive to *Staphylococcus aureus*.[65]

Rifampin 300 mg twice daily over 10–12 weeks is believed to be the best antistaphylococcal agent, in combination with clindamycin 300 mg twice daily to avoid rapid resistance. Relapse is common, and low dose of antibiotics may be necessary. Oral therapy can be combined with topical antibiotics, such as 2% mupirocin, 1% clindamycin, 1.5% fusidic acid. Topical, intralesional or systemic corticosteroids can help to reduce inflammation and pain. Oral isotretionin can be another option.[78] Successful treatment of FD with biologic therapy using adalimumab has also been described.[79]

Dissecting cellulitis Dissecting cellulitis (DC) is a rare chronic disorder of the scalp and, along with hidradenitis suppurativa (HS), acne conglobate (AC) and pilonidal cyst, it forms the follicular occlusion tetrad.[80] It usually appears in men aged 18–

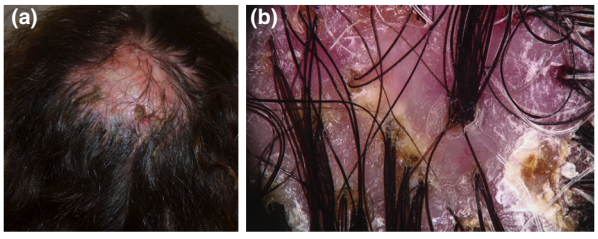


Figure 11 Clinical (a) and trichoscopic picture (b) of folliculitis decalvans.

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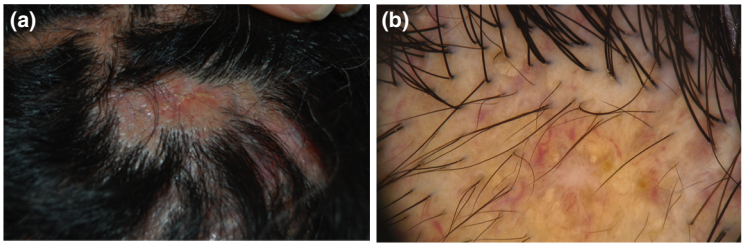


Figure 12 Clinical (a) and trichoscopic picture (b) of dissecting cellulitis.

40 years. The pathogenesis of dissecting cellulitis is currently poorly defined, although it is considered as the result of occlusion of the pilosebaceous unit, follicular rupture and inflammatory cascade.

DC manifests with perifollicular pustules, nodules, abscesses and sinuses leading to a cicatricial alopecia. In the early stage, the typical scalp lesions are perifollicular pustules, nodules and abscesses discharging seropurulent material. These nodules, filled of purulent material, are very painful and can be fluctuant, coalescencing in advanced cases with each other, showing a cerebriform aspect. Fistulae formation in very frequent, with hypertrophic or keloidal scars formation (Fig. 12).

Trichoscopic finding in DC are not specific and include discharging follicular single or grouped pustules, yellow in colour, yellowish tubular scaling, crusting and typical flame-like ecstatic vessels.

In the early stage of the disease, trichoscopy shows features of non- cicatricial alopecia with yellow structureless areas and yellow dots with a "3D" structure over dystrophic hair shaft are mainly visible. Sinus tracts and vessels named "pinpoint-like" are also present.

In advanced stages, confluent ivory-white areas with absence of follicular ostia are visible and DC is indistinguishable from cicatricial alopecia of other origin.[81]

Pathology showed that lymphocytes rather than neutrophils are the predominant inflammatory cells in the acute stage, with neutrophilic aggregates present focally around the disrupted hair follicle epithelium.

Rossi et al.[81] stated that an acneiform distension of the follicular infundibula is associated with an inflammatory process that involves also the lower portion of the terminal follicles. These hair follicles enter in the catagen/telogen phase with subsequent hair shedding.

In severe cases with nodules and fistulae, a large perifollicular and deep dermal abscesses with neutrophils and plasmacells can be observed.

Treatment is difficult and not standardized. The disease is often refractory and patients should be advised of it. Monotherapy with oral isotretinoin at 0.5–1 mg/kg/day for three months to one year has been reported as a first-line therapy. The second-line treatment include oral antibiotics, such as Clindamycin and rifampin. The third line treatment include: low dose of oral corticosteroids, carbon dioxide laser, laser epilation and radiotherapy epilation (evidence level C),[30] including 800-nm pulsed-diode laser, Nd-YAG 1064 nm, 694 nm long pulsed ruby laser. Adalimumab has been also reported to be useful for DC.[82-84]

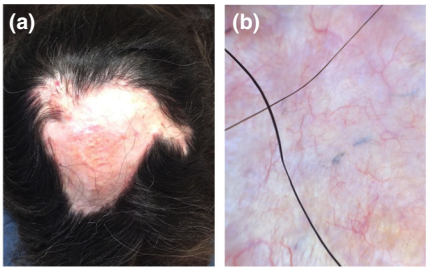


Figure 13 Clinical (a) and trichoscopic picture (b) of EPDS, with magnification of an anagen bulb visible thought the atrophic skin and telangiectasias.

Mixed inflammatory infiltrate

Erosive pustular dermatitis of the scalp Erosive pustular dermatosis of the scalp (EPDS) is a rare condition characterized by one or more areas of alopecia with erosions, thick yellow-brown crusts, superficial ulceration, and pustules.[85] EPDS occurs most commonly in the elderly with ageing of onset at 60–70 years with severe bald and sun-damaged scalp, with a previous history of trauma to the scalp.[86]

The pathology of EPDS is not fully understood: it may be correlated to traumatic event, but also an autoimmune mechanism has been proposed, based on the possible association with autoimmune disorders.[87]

The clinical presentation is frequently located at the vertex of the scalp, with sterile pustules, superficial erosions, and round or oval crusted lesions on atrophic skin.

Trichoscopic examination shows follicular keratotic plugging, milky red areas, white patches, hair shaft tortuosity, tapering hair, and absence of follicular openings that are common in

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cicatricial alopecia. With trichoscopy, it is possible to classify two different phases of EPDS: active and chronic one. The most specific feature of EPDS is the anagen bulb visible thought the atrophic skin[88] (Fig. 13). Another important aspect is the presence of prominent telangiectasias, especially after detachment of the crusts.[89]

There is no established treatment recommendation, but the topical and systemic corticosteroids are the most and first line used, in order to promote re-epithelialization.[86,90]

Conclusion

Hair disorders represent an important field of research for dermatologist and many of them are common in everyday clinical practice. The most important hair diseases are divided in non-cicatricial and cicatricial ones, and they have different clinical, dermoscopic, histological and therapeutic characteristics. Our review is focused on the fundamental principles about diagnosis and therapy, especially for the most frequent hair diseases together with the news about them with recent literature. We hope that it will be a guide for dermatologists to better diagnose and manage hair disorders.

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The patients in this manuscript have given written informed consent to the publication of their case details.

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